
CSE 457

Machine Learning

— Probabilistic Methods —
Fall 2025

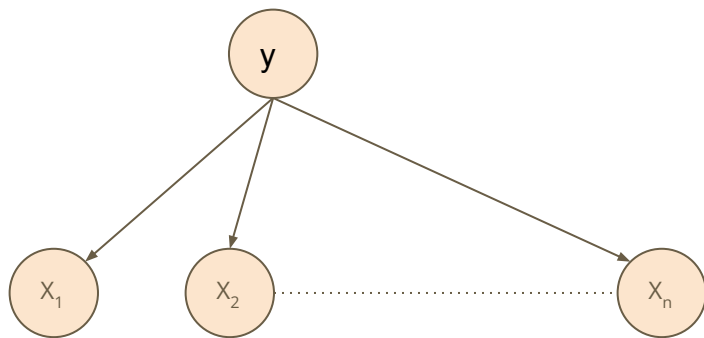
Contents

- Probabilistic Methods - Naive Bayes Classifier
 - Discrete and Continuous Cases
- How to handle non-numeric data? Mix of types?
- How to handle missing data?
- Bias vs Variance



Naive Bayes Classifier

- It works by estimating probabilities
- The prediction variable is y , and the features $X_1, X_2, \dots, X_n$
- NBC learns a Naive bayesian model
- Features in the dataset are IID, independent and identically distributed.



$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

The decision rule

Consider a binary classifier, with classes A, B (values of label, $\mathbf{y}$)

If $\Pr(y=A | X_1=v_1, X_2=v_2, \dots, X_n=v_n) > \Pr(y=B | X_1=v_1, X_2=v_2, \dots, X_n=v_n)$

Predict $y = A$

Else

Predict $y = B$

An example

$\Pr(\text{PlayGolf}=\text{yes} \mid \text{Outlook}=\text{Overcast}, \text{Temp}=\text{Mild}, \text{Humidity}=\text{Normal}, \text{Windy} = \text{True}) = ?$

$\Pr(\text{PlayGolf}=\text{no} \mid \text{Outlook}=\text{Overcast}, \text{Temp}=\text{Mild}, \text{Humidity}=\text{Normal}, \text{Windy} = \text{True}) = ?$

Predictors				Target
Outlook	Temp	Humidity	Windy	Play Golf
Rainy	Hot	High	False	No
Rainy	Hot	High	True	No
Overcast	Hot	High	False	Yes
Sunny	Mild	High	False	Yes
Sunny	Cool	Normal	False	Yes
Sunny	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Rainy	Mild	High	False	No
Rainy	Cool	Normal	False	Yes
Sunny	Mild	Normal	False	Yes
Rainy	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Sunny	Mild	High	True	No

An example

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

$\Pr(\text{PlayGolf}=\text{yes} \mid \text{Outlook}=\text{Overcast}, \text{Temp}=\text{Mild}, \text{Humidity}=\text{Normal}, \text{Windy} = \text{True}) = ?$

$= \Pr(\text{Outlook}=\text{Overcast}, \text{Temp}=\text{Mild}, \text{Humidity}=\text{Normal}, \text{Windy} = \text{True} \mid \text{PlayGolf}=\text{yes}) * \Pr(\text{PlayGolf}=\text{yes}) / \Pr(\text{Outlook}=\text{Overcast}, \text{Temp}=\text{Mild}, \text{Humidity}=\text{Normal}, \text{Windy} = \text{True})$

Predictors				Target
Outlook	Temp	Humidity	Windy	Play Golf
Rainy	Hot	High	False	No
Rainy	Hot	High	True	No
Overcast	Hot	High	False	Yes
Sunny	Mild	High	False	Yes
Sunny	Cool	Normal	False	Yes
Sunny	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Rainy	Mild	High	False	No
Rainy	Cool	Normal	False	Yes
Sunny	Mild	Normal	False	Yes
Rainy	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Sunny	Mild	High	True	No

An example

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Pr(**Outlook=Overcast**,
Temp=Mild, **Humidity=Normal**,
Windy = True | PlayGolf=yes)

Pr(PlayGolf=yes)

Pr(**Outlook=Overcast**,
Temp=Mild, **Humidity=Normal**,
Windy = True)

Predictors				Target
Outlook	Temp	Humidity	Windy	Play Golf
Rainy	Hot	High	False	No
Rainy	Hot	High	True	No
Overcast	Hot	High	False	Yes
Sunny	Mild	High	False	Yes
Sunny	Cool	Normal	False	Yes
Sunny	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Rainy	Mild	High	False	No
Rainy	Cool	Normal	False	Yes
Sunny	Mild	Normal	False	Yes
Rainy	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Sunny	Mild	High	True	No

An example

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

$\Pr(\text{Outlook}=\text{Overcast}, \text{Temp}=\text{Mild},$
 $\text{Humidity}=\text{Normal}, \text{Windy} =$
 $\text{True} \mid \text{PlayGolf}=\text{yes}) =$

$\Pr(\text{Outlook}=\text{Overcast}, \mid \text{PlayGolf}=\text{yes}) *$

$\Pr(\text{Temp}=\text{Mild} \mid \text{PlayGolf}=\text{yes}) *$

$\Pr(\text{Humidity}=\text{Normal} \mid \text{PlayGolf}=\text{yes}) *$

$\Pr(\text{Windy} = \text{True} \mid \text{PlayGolf}=\text{yes})$

Predictors				Target
Outlook	Temp	Humidity	Windy	Play Golf
Rainy	Hot	High	False	No
Rainy	Hot	High	True	No
Overcast	Hot	High	False	Yes
Sunny	Mild	High	False	Yes
Sunny	Cool	Normal	False	Yes
Sunny	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Rainy	Mild	High	False	No
Rainy	Cool	Normal	False	Yes
Sunny	Mild	Normal	False	Yes
Rainy	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Sunny	Mild	High	True	No

An example

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Pr(**Outlook=Overcast**, Temp=Mild,
Humidity=Normal, Windy =
True | PlayGolf=yes) =
(4/9) * (4/9) * (6/9)*(3/9)

Pr(**Outlook=Overcast**, | PlayGolf=yes)
= 4/9

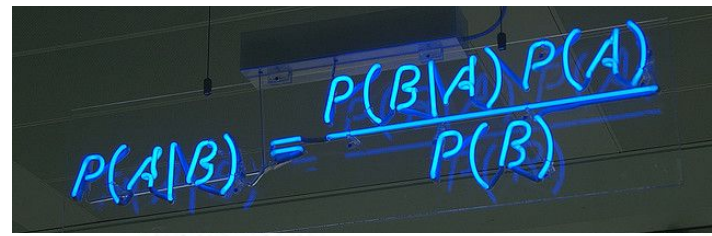
Pr(**Temp=Mild** | PlayGolf=yes)
= 4/9

Pr(**Humidity=Normal** | PlayGolf=yes)
= 6/9

Pr(**Windy = True** | PlayGolf=yes)
= 3/9

Predictors				Target
Outlook	Temp	Humidity	Windy	Play Golf
Rainy	Hot	High	False	No
Rainy	Hot	High	True	No
Overcast	Hot	High	False	Yes
Sunny	Mild	High	False	Yes
Sunny	Cool	Normal	False	Yes
Sunny	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Rainy	Mild	High	False	No
Rainy	Cool	Normal	False	Yes
Sunny	Mild	Normal	False	Yes
Rainy	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Sunny	Mild	High	True	No

An example



$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Pr(**Outlook=Overcast, Temp=Mild, Humidity=Normal, Windy = True** | PlayGolf=yes) =
 $(4/9) * (4/9) * (6/9) * (3/9)$

Pr(PlayGold=yes) = 9/14

Pr(**Outlook=Overcast, Temp=Mild, Humidity=Normal, Windy = True**) = ?

Predictors				Target
Outlook	Temp	Humidity	Windy	Play Golf
Rainy	Hot	High	False	No
Rainy	Hot	High	True	No
Overcast	Hot	High	False	Yes
Sunny	Mild	High	False	Yes
Sunny	Cool	Normal	False	Yes
Sunny	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Rainy	Mild	High	False	No
Rainy	Cool	Normal	False	Yes
Sunny	Mild	Normal	False	Yes
Rainy	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Sunny	Mild	High	True	No

An example

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

$\Pr(\text{PlayGolf=no} \mid \text{Outlook=Overcast, Temp=Mild, Humidity=Normal, Windy = True}) = ?$

$= \Pr(\text{Outlook=Overcast, Temp=Mild, Humidity=Normal, Windy = True} \mid \text{PlayGolf=no}) * \Pr(\text{PlayGolf=no}) / \Pr(\text{Outlook=Overcast, Temp=Mild, Humidity=Normal, Windy = True})$

Predictors				Target
Outlook	Temp	Humidity	Windy	Play Golf
Rainy	Hot	High	False	No
Rainy	Hot	High	True	No
Overcast	Hot	High	False	Yes
Sunny	Mild	High	False	Yes
Sunny	Cool	Normal	False	Yes
Sunny	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Rainy	Mild	High	False	No
Rainy	Cool	Normal	False	Yes
Sunny	Mild	Normal	False	Yes
Rainy	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Sunny	Mild	High	True	No

An example

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Pr(**Outlook=Overcast**, Temp=Mild,
Humidity=Normal, Windy =
True | PlayGolf=no) =
(0/5) * (2/5) * (1/5)*(3/5)

Pr(**Outlook=Overcast**, | PlayGolf=no)
= 0/5

Pr(**Temp=Mild** | PlayGolf=no)
= 2/5

Pr(**Humidity=Normal** | PlayGolf=no)
= 1/5

Pr(**Windy = True** | PlayGolf=no)
= 3/5

Predictors				Target
Outlook	Temp	Humidity	Windy	Play Golf
Rainy	Hot	High	False	No
Rainy	Hot	High	True	No
Overcast	Hot	High	False	Yes
Sunny	Mild	High	False	Yes
Sunny	Cool	Normal	False	Yes
Sunny	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Rainy	Mild	High	False	No
Rainy	Cool	Normal	False	Yes
Sunny	Mild	Normal	False	Yes
Rainy	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Sunny	Mild	High	True	No

An example

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Pr(**Outlook=Overcast, Temp=Mild, Humidity=Normal, Windy = True** | PlayGolf=no) =
(0/5) * (2/5) * (1/5)*(3/5)

Pr(PlayGold=yes) = 5/14

Pr(**Outlook=Overcast, Temp=Mild, Humidity=Normal, Windy = True**) = ?

Predictors				Target
Outlook	Temp	Humidity	Windy	Play Golf
Rainy	Hot	High	False	No
Rainy	Hot	High	True	No
Overcast	Hot	High	False	Yes
Sunny	Mild	High	False	Yes
Sunny	Cool	Normal	False	Yes
Sunny	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Rainy	Mild	High	False	No
Rainy	Cool	Normal	False	Yes
Sunny	Mild	Normal	False	Yes
Rainy	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Sunny	Mild	High	True	No

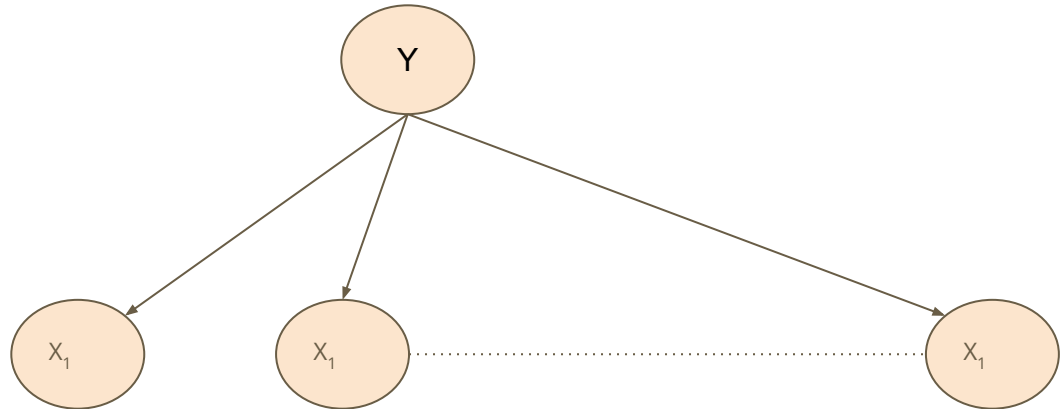
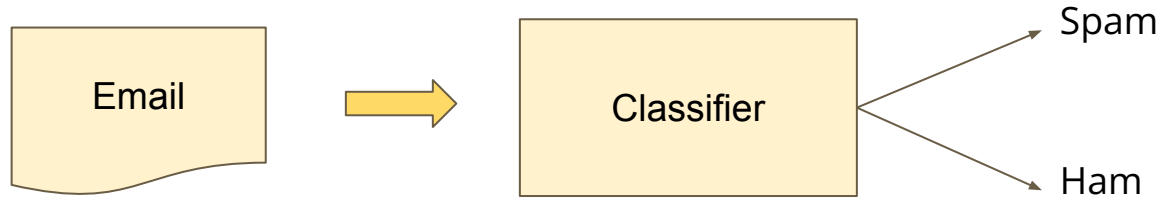
Pros and Cons

- Easy to implement
- Training faster, no gradient descent needed
- Does not overfit
- Less memory and CPU requirement
- Easy to retrain with new data
- Handles both numeric and categorical data
- Handles missing data automatically

- Ensembling, bagging, boosting does not work, no variance to reduce
- Zero frequency problem for categorical data
- IID assumption - often not practical
- Numerical underflow
- Skewed data causes problems in prior calculation

Another Example - Text Classification

- Spam Filter
- Bag of words



Supervised Learning Problem

- Bag of words model
- Features are frequency of the words
- Maintains a dictionary of words == vocabulary

SPAM

OFFER IS SECRET
SECRET SPORTS LINK
CLICK SECRET LINK

HAM

WENT PLAY SPORTS
PLAY SPORTS TODAY
SECRET SPORTS EVENT
SPORTS IS TODAY
SPORTS COSTS MONEY

Email Filter

- Vocabulary = 12
- $P(\text{SPAM}) = ?$
- $P(\text{"SECRET"} \mid \text{SPAM}) = ?$
- $P(\text{"SECRET"} \mid \text{HAM}) = ?$
- $P(\text{"SPORTS"} \mid \text{SPAM}) = ?$
- $P(\text{"SPORTS"} \mid \text{HAM}) = ?$

SPAM

OFFER IS SECRET
SECRET SPORTS LINK
CLICK SECRET LINK

HAM

WENT PLAY SPORTS
PLAY SPORTS TODAY
SECRET SPORTS EVENT
SPORTS IS TODAY
SPORTS COSTS MONEY

Email Filter

Message = "SPORTS"

SPAM

OFFER IS SECRET
SECRET **SPORTS** LINK
CLICK SECRET LINK

HAM

WENT PLAY SPORTS
PLAY SPORTS **TODAY**
SECRET SPORTS **EVENT**
SPORTS IS TODAY
SPORTS **COSTS MONEY**

If $\Pr(\text{SPAM} \mid \text{Message} = \text{"SPORTS"}) > \Pr(\text{HAM} \mid \text{Message} = \text{"SPORTS"})$

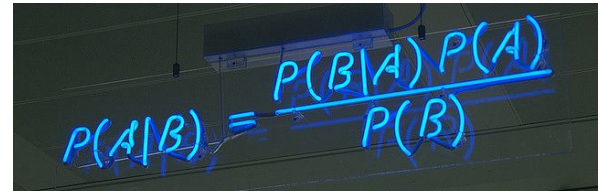
Message = **SPAM**

Else

Message = **HAM**

$\Pr(\text{SPAM} \mid \text{SPORTS}) = \Pr(\text{SPORTS} \mid \text{SPAM}) * \Pr(\text{SPAM}) / \Pr(\text{SPORTS})$

$\Pr(\text{HAM} \mid \text{SPORTS}) = \Pr(\text{SPORTS} \mid \text{HAM}) * \Pr(\text{HAM}) / \Pr(\text{SPORTS})$



A photograph of a chalkboard with the formula for Bayes' theorem written in blue chalk. The formula is $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$. The variables A and B are written in a way that suggests they are being defined or substituted, with A likely representing the event of interest (like SPAM) and B representing the evidence (like the word SPORTS).

Email Filter

Message = "SECRET IS SECRET"

SPAM

OFFER IS SECRET
SECRET **SPORTS** LINK
CLICK SECRET LINK

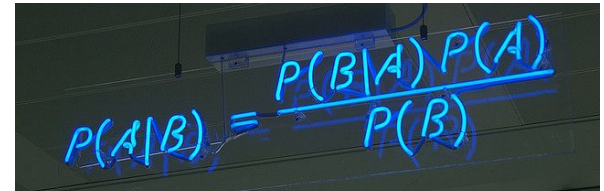
HAM

WENT PLAY SPORTS
PLAY SPORTS **TODAY**
SECRET SPORTS **EVENT**
SPORTS IS TODAY
SPORTS **COSTS MONEY**

$\Pr(\text{SPAM} \mid \text{MESSAGE}) = \Pr(\text{MESSAGE} \mid \text{SPAM}) * \Pr(\text{SPAM}) / \Pr(\text{MESSAGE})$

$\Pr(\text{HAM} \mid \text{MESSAGE}) = \Pr(\text{MESSAGE} \mid \text{HAM}) * \Pr(\text{HAM}) / \Pr(\text{MESSAGE})$

$\Pr(\text{MESSAGE}) = \Pr(\text{MESSAGE} \mid \text{SPAM}) * \Pr(\text{SPAM}) + \Pr(\text{MESSAGE} \mid \text{HAM}) * \Pr(\text{HAM})$



A photograph of a chalkboard with the formula for Bayes' theorem written in blue chalk. The formula is $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$. The variables A and B are written in a stylized, slightly slanted font.

Email Filter

Message = "TODAY IS SECRET"

SPAM

OFFER IS SECRET
SECRET SPORTS LINK
CLICK SECRET LINK

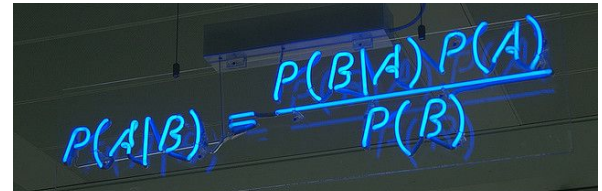
HAM

WENT PLAY SPORTS
PLAY SPORTS TODAY
SECRET SPORTS EVENT
SPORTS IS TODAY
SPORTS COSTS MONEY

$\Pr(\text{SPAM} \mid \text{MESSAGE}) = \Pr(\text{MESSAGE} \mid \text{SPAM}) * \Pr(\text{SPAM}) / \Pr(\text{MESSAGE})$

$\Pr(\text{HAM} \mid \text{MESSAGE}) = \Pr(\text{MESSAGE} \mid \text{HAM}) * \Pr(\text{HAM}) / \Pr(\text{MESSAGE})$

$\Pr(\text{MESSAGE}) = \Pr(\text{MESSAGE} \mid \text{SPAM}) * \Pr(\text{SPAM}) + \Pr(\text{MESSAGE} \mid \text{HAM}) * \Pr(\text{HAM})$



A photograph of a chalkboard with the formula for Bayes' theorem written in blue chalk. The formula is $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$. The variables A and B are underlined in the original image.

Laplacian Smoothing

- $\Pr(X) = \text{Count}(x) / \text{Total Count}$
- In Laplacian Smoothing,
 - $\Pr(X) = (\text{Count}(X) + k) / (\text{Total Count} + K * |x|)$
- $P(\text{SPAM}) = ?$
- $P(\text{"TODAY"} \mid \text{SPAM}) = ?$
- $P(\text{"IS"} \mid \text{SPAM}) = ?$
- $P(\text{"SECRET"} \mid \text{SPAM}) = ?$
- $P(\text{"TODAY"} \mid \text{HAM}) = ?$
- $P(\text{"IS"} \mid \text{HAM}) = ?$
- $P(\text{"SECRET"} \mid \text{HAM}) = ?$

SPAM

OFFER IS SECRET
SECRET SPORTS LINK
CLICK SECRET LINK

HAM

WENT PLAY SPORTS
PLAY SPORTS TODAY
SECRET SPORTS EVENT
SPORTS IS TODAY
SPORTS COSTS MONEY

Message = "TODAY IS SECRET"

Advanced Email Filters

- Known spamming IP
- Have you emailed the person before?
- Other people received the same message?
- Email header consistent?
- ALL CAPS?
- URLs are pointing correctly?
- Are addressed by your correct name?

Digit Recognition

- Features = Pixels
- Class Labels = 0,1,...,9
- Lets use sklearn



Gaussian Naive Bayes

- How to handle numeric data?
- We assume that these values are sampled from a gaussian distribution

$$P(x_i | y) = \frac{1}{\sqrt{2\pi\sigma_y^2}} \exp\left(-\frac{(x_i - \mu_y)^2}{2\sigma_y^2}\right)$$

$$P(\text{Income} = 120 | \text{Evade} = \text{No}) = \frac{1}{\sqrt{2 \times \pi \times 2975}} \exp\left(-\frac{(120 - 110)^2}{2 \times 2975}\right)$$

$$P(\text{Income} = 120 | \text{Evade} = \text{Yes}) = \frac{1}{\sqrt{2 \times \pi \times 25}} \exp\left(-\frac{(120 - 90)^2}{2 \times 25}\right)$$

<i>Tid</i>	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Bias vs Variance

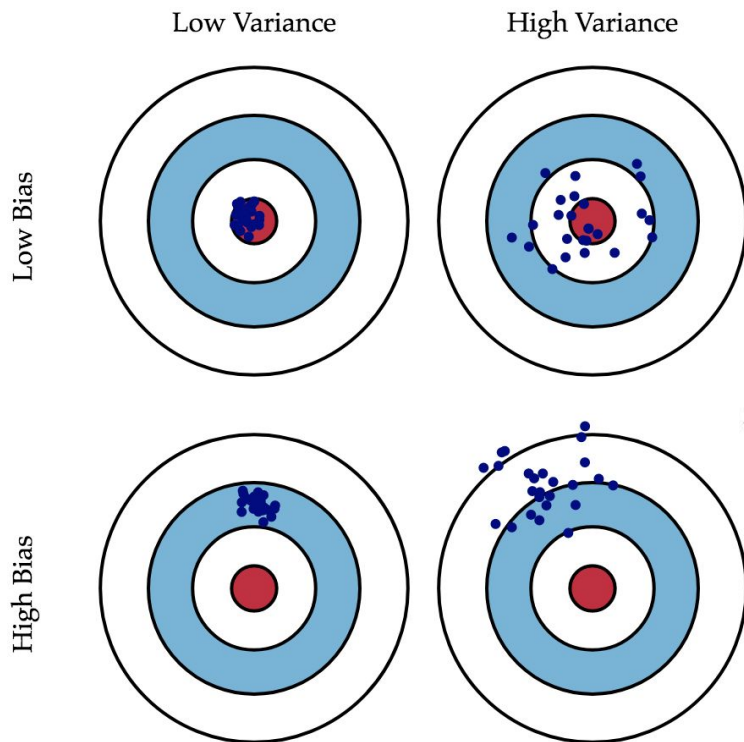
Voting Republican	Voting Democratic	Non-Respondent	Total
13	16	21	50

- Suppose you want to find the number of votes that Joe Biden will get in different states
 - The real value is y
 - Your prediction is $\hat{f}(x)$
 - The bias is the difference
- Variance is the expectation of the squared deviation of a random variable from its mean
 - Estimate of the target function will change if different training data was used.

$$bias = E[\hat{f}(x)] - f(x)$$

$$var(x) = E[(\hat{f}(x) - E[\hat{f}(x)])^2]$$

Bias vs Variance



$$Err(x) = Bias^2 + Variance + Irreducible\ Error$$

Bias vs Variance

Low Bias

Depends on Training Data,
Not much assumption on
the model, KNN, SVM,
Decision Tree

High Bias

Assumptions on the
model, underfitting, not
adequate data, linear and
logistic regression

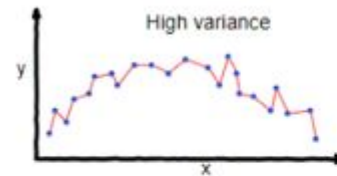
+

Low Variance

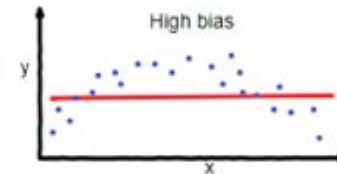
Changing the dataset
makes small changes
on the model

High Variance

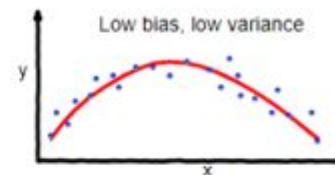
Depends on training
data, captures noise,
overfits



overfitting

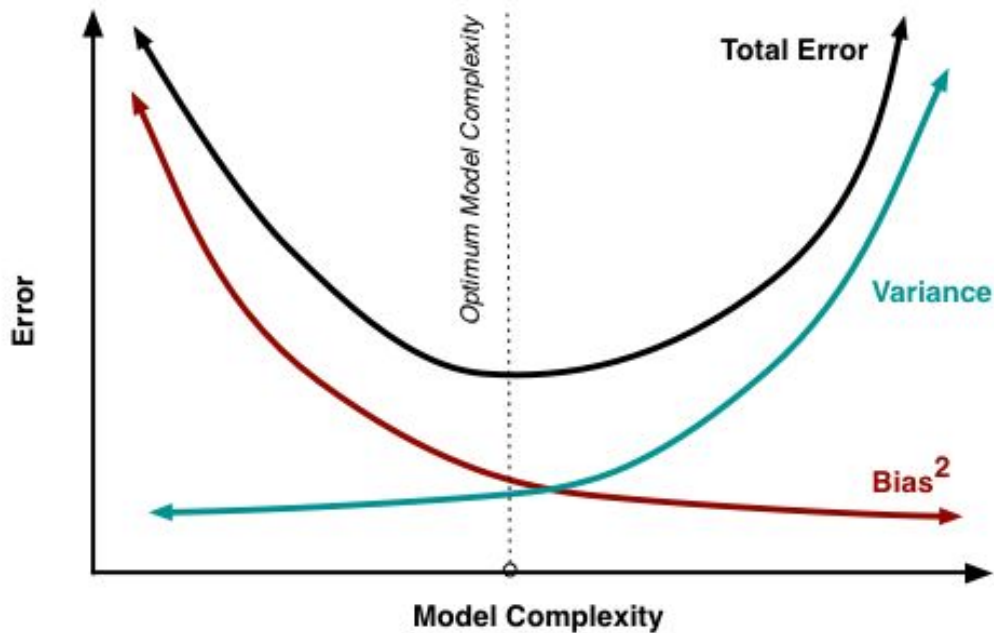


underfitting



Good balance

Bias vs Variance



$$Err(x) = \text{Bias}^2 + \text{Variance} + \text{Irreducible Error}$$